

Four days of fever in a 10-month-old girl

What the orders and results showed, and what to carry to the next infant with fever and no source.

CASE IN BRIEF

Earlier, you took a phone call from the parent of a previously healthy, immunized 10-month-old girl with four days of fever near 39 C, fussiness that cleared after acetaminophen, and no other symptoms. The parent asked about Tylenol and was reluctant to bring her in because of work. The aim of the call was a fever-without-source assessment, an office visit within 24 hours, and a shared plan. You debriefed that call. In this final step you examined Megan, placed orders in a simulator, and read the results. She went from fever without a source to a confirmed febrile urinary tract infection. You then debriefed the case with an attending, who asked how you would diagnose, treat, and follow her.

What each test was for

TEST	WHAT IT WOULD CHANGE	RESULT	WHAT WE DID
Catheter urinalysis, Gram stain and culture, collected before any antibiotic	Confirms or excludes UTI. The culture, read with the urinalysis, makes the diagnosis.	LE 3+, nitrite positive, many WBC and bacteria. Gram-negative rods. <i>E. coli</i> over 100,000 CFU/mL at 48 h.	Diagnosed a febrile UTI, then started an antibiotic.
Blood count, CRP, procalcitonin	Adds weight toward kidney involvement. Does not change the first step.	WBC 16.4, CRP 4.1 mg/dL, procalcitonin 2.4 ng/mL.	Read as supporting, not proving, deeper infection. No cutoff exists for this age.
Basic metabolic panel and hydration check	Decides whether she can be managed by mouth.	Na 138, K 4.2, HCO3 20, BUN 14, Cr 0.3. Mucous membranes slightly dry. Took 60 mL of formula and kept it down.	Oral therapy was reasonable because she could keep fluids down.
Blood culture and lumbar puncture	Only matters if the result would change the plan.	Not available in this simulation.	A judgment call. Ask what result would change your plan before ordering.

If it had come back the other way

She was sent home and no urine was obtained	The source of the fever is never found and she returns still febrile. In a febrile infant with no other source, test the urine before she leaves.
Only a bag urine was sent, and it grew mixed organisms	A bag specimen can reflect contamination and does not establish the diagnosis. In a child who is not toilet trained, collect by catheter or suprapubic aspiration.
Normal urinalysis and fever on day six	The urine no longer explains the fever. Go back to the exam and the differential rather than stopping the search.
She vomits the first dose and will not take fluids	Oral therapy is as effective as short-course IV only when she can keep it down. Reconsider route and place of care.

Three take-home points

CARRY IT FORWARD

The next time one detail changes, such as a younger infant, a bag specimen, a normal urinalysis or a vomited dose, say what changes in your plan and what does not.

- 1 Always consider a UTI in a febrile young diapered child when no other source is found.** Infants do not report dysuria and cannot localize flank pain, so the absence of urinary symptoms or costovertebral angle tenderness does not argue against it.
- 2 A urinalysis with white cells is not always a UTI.** The diagnosis needs pyuria and a single uropathogen at or above 50,000 CFU/mL from a catheter or suprapubic specimen collected before the first dose. A positive bag or a urinalysis alone does not meet that standard.
- 3 Treat promptly, dose by weight, and let how well she is and whether she keeps it down choose the route.** A well, stable child who retains fluids and medication can be treated by mouth. Concern for sepsis or hemodynamic instability, a toxic appearance, or vomiting means IV therapy and admission. Expect improvement in 48 to 72 hours, and plan what you will do if it does not come.

References

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